

Raksha Ramesh

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EDUCATION

University of Pennsylvania

Master of Science in Engineering in Computer and Information Science

Major: Computer and Information Science; *Concentration:* Artificial Intelligence

Philadelphia, PA

Dec 2025

GPA: 3.88/4.0

PES University

Bachelor of Technology in Computer Science and Engineering

Specialization: Machine Intelligence and Data Science

Bangalore, India

May 2022

GPA: 9.15/10

WORK EXPERIENCE

Crum & Forster

Data Science Intern

Philadelphia, PA

June 2025 – Sept 2025

- Built an **OCR pipeline (93% accuracy, 320+ apps/month)** consulting with senior stakeholders, cutting review time by **~10 minutes/file** and speeding underwriting by **20%**.
- **Decommissioned Pony ORM in 10 production microservices**, implementing SQL context managers that eliminated connection issues, **boosted performance by 20%**, and improved reliability for **50+ business users**.
- Created an automated **API validation tool (10 services, 500+ payloads)** that reduced testing by **~30 minutes/service**, standardized edge-case coverage, and earned team-wide recognition.

Walmart Global Tech

Software Development Engineer II

Bangalore, India

July 2022 – May 2023

- **Maintained 5 critical microservices** for Walmart US e-commerce, ensuring 2M+ daily transactions ran with 99.9% availability.
- Led **rate limiting** rollout validated with peak-traffic simulations and **architect reviews**; first to implement, **cutting outages by 15% (~\$15M)** and publishing a blueprint **adopted org-wide**.
- Spearheaded integration of OpenTelemetry **distributed tracing** into service workflows, **reducing incident resolution time by 20%** and enabling faster root-cause analysis across the team.

Software Development Intern

Jan 2022 – June 2022

- Designed and built end-to-end integration of PowerSport Category vehicles into product catalogs, adding **1,200+ products** and enabling targeted filtering that **improved discoverability by 25%**.
- Built and maintained **Grafana and Splunk dashboards** for 5 microservices, **reducing detection time** for service issues by **40%** and improving uptime reliability.
- Orchestrated **testing and deployment** of 2 production services, **aligning QA, product, and engineering** teams to ensure seamless launches with minimal downtime.

Ellipses Innovation

Marketing Intern

Bangalore, India

May 2021 – Aug 2021

- Developed data-driven **Instagram marketing strategy** by analyzing engagement metrics and audience demographics — increasing follower growth by 35% and generating the company's first recurring sales (**20+ orders/week**).
 - Conducted **competitor benchmarking** and **market research** to refine content mix, posting frequency, and optimal posting times, improving campaign engagement rates.
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PROJECTS AND PUBLICATIONS

AI/ML & NLP PROJECTS

Segment Based Abnormality Detection in EEG Recordings

Published in IEEE, Aug 2022

- Achieved 90.78% **classification** accuracy (surpassing SOTA) for **EEG abnormalities** using Empirical Wavelet Transform (EWT), Recursive Feature Elimination (RFE), and Linear SVM.
- Evaluated multiple **ML pipelines** (EMD, EWT, k-NN, SVM, XGBoost, MLP) and optimized performance through **iterative experimentation**.

DreamBook

Oct 2024 – Dec 2024

- **Designed an AI diary platform** that digitized handwritten, audio, and text entries using Google Cloud Vision, Speech-to-Text, and Stable Diffusion (DreamBooth + LoRA), generating consistent comic-style illustrations.
- **Deployed via Gradio** for seamless uploads and real-time AI-generated outputs.

BART-to-Edge

Oct 2024 – Dec 2024

- **Optimized mBART/M2M100 with LoRA + layer freezing**, reducing GPU memory usage by 50% and training time by 30%.
- **Enabled sub-1B parameter real-time Chinese–English translation** on edge devices while maintaining BLEU score parity.

Lexical Simplification

Feb 2021 – May 2021

- Developed contextual word substitution pipeline improving text accessibility for diverse readers — reducing reading complexity scores by 20% and improving comprehension in testing.
- Used NLTK, gensim, and pandas for tokenization, synonym generation, and semantic ranking with part-of-speech tagging and frequency-based filtering.

Deep Learning for Text Summarization

Feb 2020 – May 2020

- Implemented both extractive and abstractive summarization pipelines to improve readability and reduce redundancy in generated summaries.
- Benchmarked with ROUGE and BLEU, showing abstractive methods improved semantic coherence while reducing repetition by 25%.

DATA SCIENCE & ANALYTICS PROJECTS

Optimizing ANN for Learned Cardinalities

Oct 2023 – Dec 2023

- Tuned MSCN-based learned cardinality estimator in PyTorch, adjusting encoder–decoder MLP architecture for improved accuracy.
- Selected Adam + MLSE loss for smoother convergence, lowering median Q-error and standardizing experiment workflows.

World Development Indicators Analysis

Aug 2020 – Dec 2020

- Modeled macroeconomic growth using regression on World Bank WDI data, achieving RMSE of 2.7% (developed) and 4.3% (developing).
- Analyzed sector drivers to produce actionable contrasts for policy-focused interpretations.

YACS (Yet Another Centralized Scheduler)

Aug 2020 – Dec 2020

- Built master–worker scheduler in Python with sockets + multithreading, implementing Random, Round Robin, and Least-Loaded scheduling policies.
- Added map-reduce task dependencies and back-pressure controls to reduce idle CPU time.

SYSTEMS & INFRASTRUCTURE PROJECTS

PennOS – Custom Operating System

Jan 2025 – May 2025

- Developed UNIX-like OS in C with FAT filesystem, multi-level feedback queue scheduler, and preemptive user-space threading.
- Created custom shell with built-in commands, process control syscalls, and concurrency-safe inter-thread communication.

Mini-Compiler for JavaScript

Aug 2020 – Dec 2020

- Created a compiler in C implementing **lexical analysis**, **syntax parsing**, and **semantic analysis** for a JavaScript subset.
- Generated optimized **three-address code** with constant folding and dead code elimination before emitting assembly-like output.
- Designed a **symbol table** with scope management and type-checking for accurate semantic validation.

Elevator Control System

Jan 2020 – May 2020

- Designed and implemented a C-based elevator control simulation using an **ordered doubly linked list** to maintain pending floor requests in priority order.
- Simulated real-time operations with direction-aware scheduling to reduce average wait times.

LEADERSHIP AND EXTRACURRICULARS

University of Pennsylvania

Philadelphia, PA

Head Teaching Assistant/ Product Manager

Aug 2024 – Present

- Directed product strategy for 5 cross-functional student teams, leading MVP development cycles through agile planning, user feedback loops, and stakeholder demos.
- Coordinated a team of 11 TAs, overseeing training, task allocation, and day-to-day operations to ensure smooth course delivery.
- Managed all course logistics — scheduling, grading workflows, resource planning — to support 100+ students and faculty.
- Conducted user research and usability testing to refine requirements, improving deliverable adoption rates.
- Managed timelines, budgets, and weekly stand-ups, evaluating 20+ product demos for quality and market fit.

U&I Trust

Bangalore, India

Chair – ACM-W - Student Chapter

May 2022 – Jul 2023

- Directed the English literacy program for **20+ underprivileged girls**, designing custom lesson plans to improve conversational and written fluency.
- Led volunteer recruitment drives, growing the teaching staff by 50% to meet demand.
- Spearheaded fundraising initiatives that fully covered annual program costs.

PES University

Bangalore, India

Chair – ACM-W - Student Chapter

Aug 2020 – Jul 2022

- Expanded chapter membership 4× through inclusive outreach and mentorship programs.
- Secured corporate sponsorships and VC funding for Women's Day Ideathon, supporting top 3 winning ideas.
- Organized 15+ workshops, hackathons, and speaker events promoting diversity in tech.

Event Lead / Organizer

Aug 2019 – Jul 2022

- Planned and executed national-level **technical and cultural festivals** attracting **10,000+ attendees** annually.
- Managed logistics, sponsorship acquisition, and vendor negotiations to run events on budget.
- Designed event schedules and marketing campaigns, increasing footfall by ~20% year-over-year.

SKILLS & INTERESTS

- **Programming:** Python, TypeScript, Node.js, C, GraphQL, SQL, Git, Postman, CI/CD, JIRA, Confluence, Tableau
- **AI & Machine Learning:** PyTorch, NumPy, pandas, scikit-learn, Empirical Wavelet Transform (EWT), Empirical Mode Decomposition (EMD), Recursive Feature Elimination (RFE), LoRA, DreamBooth, Stable Diffusion, BLEU, BERTScore
- **NLP & Vision:** LLMs, mBART, M2M100, NLTK, Google Cloud Vision, Speech-to-Text, Text/Image Classification
- **Cloud & Infrastructure:** AWS (CDK, Step Functions, Bedrock, Textract, BDA, Serverless Framework), Google Cloud (Vision, Speech-to-Text), Kubernetes, Istio
- **Data Engineering & Observability:** Pony ORM, Grafana, Splunk, Prometheus, OpenTelemetry, Jaeger
- **Systems & Compilers:** FAT filesystem, multi-level feedback queue scheduler, concurrency control, compiler design (lexical/syntax/semantic analysis, three-address code generation)
- **Product & Collaboration:** Agile/Scrum, MVP development, user research, A/B testing